

Certified Perceptual Shielding for Adaptive Bitrate Streaming

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Abstract

Client-side adaptive bitrate (ABR) controllers—whether heuristic, model-based, or learned—share a blind spot that surfaces chunk by chunk: a request can become unsafe when throughput fluctuates, or wasteful once the rate–quality ladder has saturated in Video Multimethod Assessment Fusion (VMAF). We address both failure modes with a *Certified Perceptual Shield* (CPS), a model-agnostic runtime projection built from three components. *VMAF-knee bandwidth banking* lowers even feasible proposals to the smallest rung whose VMAF lies within a perceptual budget ε , converting saturation into banked buffer with negligible quality loss. An online *conformal* lower bound on throughput, calibrated with a finite-sample $(n+1)$ rank correction, yields distribution-free coverage $1-\alpha$ under exchangeability. *Certified feasibility* then rejects any download whose time under that bound would exceed the buffer margin, yielding a per-chunk *probabilistic* stall certificate under the conformal assumption—a coverage guarantee, not a promise of zero stalls. We evaluate CPS on 204 paired synthetic 5G episodes over twelve reference titles chosen for *ladder diversity*, from saturation-rich animation to steep natural motion. Against a conformal safety-only baseline, banking significantly reduces delivered bitrate and rebuffering on every deterministic host we test (greedy through RobustMPC: $-4.5\%/-9.3\%$ to $-3.8\%/-9.2\%$), with mean VMAF differences within ± 1.0 (TOST) and empirical coverage at 0.919 against a 0.90 target. Because effect sizes follow top-rung saturation, the diversified roster yields conservative but credible savings. A risk-aware, predictive variant deepens banking ahead of forecast throughput dips (greedy: -14.6% bitrate, -12.6% rebuffering). As a

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supporting result, co-training a PPO policy with the shield in the loop shifts the certified operating point toward higher fidelity (+2.93 mean VMAF at +0.55 s added rebuffering vs. shield-at-eval-only), with QoE crossover $w^*=5.31$ VMAF points per second.

Keywords: Adaptive Bitrate Streaming, Runtime Shielding, VMAF, Conformal Prediction, Certified Safety, Policy–Shield Co-design, Video QoE

1. Introduction

Client-side adaptive bitrate (ABR) policies optimize average quality of experience (QoE) through buffer heuristics [1], model-predictive control [2], and deep reinforcement learning (DRL) [3, 4]. Session averages, however, conceal per-chunk risk: a single segment can still stall under volatile sub-6 GHz and 5G/mmWave throughput [5–7], while a comfortable buffer invites the opposite inefficiency. Commercial rate ladders often *saturate* in Video Multimethod Assessment Fusion (VMAF) [8]—top rungs consume far more bytes yet add little perceptual quality—so requesting them when the buffer is healthy burns bandwidth that could cushion the next dip.

A runtime *shield* sits between the base policy and the player, projecting each proposal before download [9]. The standard repair decrements the representation index until a buffer-feasibility test passes. That repair is content-blind: it ignores VMAF saturation, never banks bandwidth *proactively*, and reduces throughput pessimism to a hand-tuned factor with no coverage guarantee. Deployed ladders carry exploitable structure—rate–quality saturation near the top—that enables perceptually near-lossless downshifts and frees buffer headroom.

We introduce a *Certified Perceptual Shield* (CPS) that targets this structure while stating a formal coverage guarantee:

1. **VMAF-knee bandwidth banking.** Even a feasible action is lowered to the smallest rung j whose VMAF lies within a budget ε of the proposal; the downshift is perceptually bounded, and reclaimed bytes bank as buffer for later chunks.
2. **Conformal throughput lower bound.** Download-time tests use an online split-conformal lower bound on throughput with target coverage $1-\alpha$, calibrated from residual ratios with a finite-sample $(n+1)$ rank correction [10, 11].
3. **Certified feasibility.** After banking, CPS admits no rung whose download time under the lower bound exceeds the buffer minus a safety margin,

yielding a per-chunk *probabilistic* stall certificate with coverage $1-\alpha$ under residual exchangeability rather than a deterministic no-stall guarantee (Proposition 1).

Prior VMAF-aware *ranking* shields fare no better on monotone ladders: maximizing retained VMAF among feasible indices collapses to highest-feasible-index projection (Proposition 2), so a perceptual-loss budget cannot constrain choices when VMAF is nondecreasing in the rung index. The wrapper is host-agnostic: the same projection attaches to bitrate-greedy, buffer-based (BBA), and learned PPO controllers alike. To separate the two mechanisms, we run three paired arms on identical episode seeds: RAW (unshielded), SAFETY (conformal feasibility only), and CERTIFIED (feasibility plus banking). Banking then isolates to the CERTIFIED–SAFETY contrast, and safety to SAFETY/CERTIFIED vs. RAW.

We evaluate these claims on 204 paired synthetic 5G episodes over twelve ladder-diverse titles with $\varepsilon=1.0$ and $\alpha=0.10$. Certified banking against safety-only significantly reduces mean bitrate and rebuffering on greedy, BBA, BOLA, and RobustMPC (Table 5; e.g. $-4.5\%/-9.3\%$ for greedy, paired Wilcoxon $p=8.5\times 10^{-4}$). Magnitudes cluster near $\pm 4\%/\pm 9\%$ because steep, non-saturated ladders leave fewer near-lossless downshifts, yet the direction holds everywhere. Mean VMAF differences remain within $\pm\varepsilon$ by TOST equivalence, with empirical conformal coverage at 0.919 against a 0.90 target. A risk-aware, predictive extension widens the perceptual budget when the buffer—or a short look-ahead under a throughput floor—enters a danger zone (greedy: -14.6% bitrate, -12.6% rebuffering). On source-disjoint *real-measured* broadband traces (FCC and Norway/Nor-net), banking headroom collapses (-0.9% bitrate) while safety still cuts stalls (-98.7% vs. raw greedy), delineating when perceptual banking pays off. Co-training PPO with CPS in the loop (a supporting result) shifts the certified point toward higher fidelity ($+2.93$ mean VMAF at $+0.55$ s rebuffering; Table 6), with crossover $w^*=5.31$ VMAF points per second of stall ($p=5.0\times 10^{-5}$ for VMAF).

The main contributions are:

- **Certified Perceptual Shield.** A model-agnostic runtime projection combining VMAF-knee bandwidth banking, a conformal throughput lower bound with finite-sample coverage, and certified buffer feasibility (Section 4).
- **Isolated banking evaluation.** Paired RAW/SAFETY/CERTIFIED rollouts on greedy, BBA, BOLA, RobustMPC, and PPO hosts, with Wilcoxon tests on bitrate and rebuffering and TOST non-inferiority on VMAF within $\pm\varepsilon$ (Section 5).

- **Risk-aware, predictive banking.** A buffer- and look-ahead-conditioned perceptual budget that pre-banks before forecast dips, improving the bandwidth–stall trade-off under the same ε ceiling at comfortable buffers (Section 4.3).
- **Regime characterization.** We show that banking gains concentrate where top rungs are often feasible (synthetic 5G moderate buffer), whereas real-trace broadband stress is dominated by safety projection (Section 5).
- **Policy–shield co-design.** Training with CPS in the loop shifts the certified operating point toward higher fidelity; we report the QoE crossover weight w^* that separates fidelity-favoring from stall-averse regimes (Section 5.4).

The remainder of this paper is organized as follows. Section 2 reviews related work, and Section 3 formalizes the streaming model. Section 4 presents CPS and optional co-training. Section 5 reports the evaluation, Section 6 discusses limitations, and Section 7 concludes.

2. Related work

Three areas bear on client-side representation selection: model-based ABR, learned ABR, and constrained control with runtime shielding. Our focus is the control policy and its runtime action projection; throughput prediction lies outside this scope. Predictors like CS2P [12] play a complementary role, feeding inputs to model-based or learned controllers [2, 3]. ABR surveys [7, 13, 14] and 5G/mmWave measurements [5, 15] converge on one empirical lesson: control must remain robust when throughput fluctuates sharply.

2.1. Heuristic and model-based ABR

MPEG Dynamic Adaptive Streaming over HTTP (DASH) separates packaging from transport, so clients run local policies over ladder metadata and download observations [16]. Buffer-threshold heuristics such as BBA [1] remain simple and interpretable, whereas BOLA [17] derives its decisions from a Lyapunov-based utility formulation. RobustMPC [2] plans ahead, picking representations by solving a short-horizon model predictive control (MPC) problem under estimated future throughput. The payoff is interpretable thresholds and horizons; the cost is fragility, since performance depends on manual calibration and on prediction accuracy that varies across networks and devices [5, 6]. Moreover, conventional feasibility rules rarely exploit VMAF saturation to bank bandwidth in advance [8]. Adjacent work targets other stages of delivery: SODA [18] pursues smoothness-aware online

control, while edge assistance [19] and Oboe-style automatic tuning [20] address separate parts of the system.

2.2. Deep reinforcement learning for ABR

Pensieve [3] established the template: a trace-driven DRL formulation for ABR, followed by D-DASH [21] and more recent PPO-based controllers [22]. Their neural policies combine throughput histories, buffer state, and manifest features [23]. Later systems built on this line: Comyco [24] trains by imitation against a VMAF-based QoE objective, showing that perceptual signals help inside the learned policy. Loki [25] combines learned features with a rule-based model, reducing long-tail failures in production. Fugu [26] deploys learned control in live networks and exposes the distribution shift between simulation and deployment. Taken together, these systems motivate constrained learning and decision-time enforcement, because an expected-return objective alone encodes no explicit per-episode stall or smoothness budget [27, 28]. Explicit constraints do not fully close this gap: a controller trained under a CMDP retains guarantees only over the sampled trajectory distribution, so an unseen blockage pattern can still induce an unsafe proposal at a single chunk boundary. This gap argues for splitting the stack into a learned performance layer and a runtime enforcement layer: additional training cannot substitute for runtime validation.

2.3. Safe/constrained RL, shields, and the content-blindness gap

CMDPs and Lagrangian primal–dual methods optimize expected return under explicit constraints [29–31]. Safe-RL surveys catalogue related families: shielding, Lyapunov, and risk-sensitive schemes [28, 32]. COREL [27] is closest to our training formulation. It casts ABR over commercial 5G/mmWave traces as a CMDP with a stall-time constraint, solved by a primal–dual actor–critic. That constraint, however, operates within training; it is not a standalone runtime projection with intervention telemetry. Runtime shielding takes the opposite approach: it repairs a proposed action before execution and can log each intervention [9]. Surveys split the family into *pre-shielding*, which prunes the action set, and *post-shielding*, which overwrites an unsafe proposal [33]. By that taxonomy, CPS is a post-shield whose safety set is drawn by a conformal feasibility test rather than a temporal-logic model.

This distinction shapes our design: we separate an optional learned performance layer from a deterministic runtime CPS. Learned policies such as Lagrangian PPO can pursue expected stall and smoothness budgets across training rollouts [27, 30], but they cannot certify per-chunk feasibility under conformal throughput

bounds. CPS targets exactly this gap: before every download it (i) banks bandwidth on VMAF-saturated rungs within a perceptual budget ε , and (ii) projects onto actions feasible under a distribution-free throughput lower bound with coverage $1-\alpha$ [9–11]. Its use of perceptual data also differs from Comyco, where VMAF enters only the learned objective [24]: CPS reads the ladder at decision time for *proactive* banking, rather than only emergency index repair. Table 1 summarizes this positioning.

Table 1: Positioning of client-side ABR methods by training constraints, deterministic runtime projection, conformal throughput bounds, and VMAF-knee banking.

Work	Year	Paradigm	Constr.	Guard	Conf.	Bank
BBA [1]	2014	Heuristic	N	N	N	N
BOLA [17]	2018	Lyapunov util.	Imp.	N	N	N
RobustMPC [2]	2015	MPC/lookahead	N	N	N	N
Pensieve [3]	2017	DRL	N	N	N	N
Comyco [24]	2019	Quality-aware IL	N	N	N	N
Loki [25]	2021	Rule–RL fusion	N	$\approx$	N	N
COREL [27]	2023	Constr. DRL	Y	N	N	N
SODA [18]	2024	Online control	N	N	N	N
Ours (CPS)	—	Shield ($\pm$ RL)	$\approx$	Y	Y	Y

Constr.: explicit stall budget in training (Y), implicit (Imp.), optional/partial ($\approx$), or none (N). **Guard:** auditable action projection before download [9].

Conf.: conformal throughput lower bound with stated coverage [10]. **Bank:** VMAF-knee bandwidth banking within a perceptual budget ε .

3. System model and problem formulation

Table 2 summarizes the notation used throughout.

Table 2: Summary of notation.

Symbol	Meaning
k, T_c, K_e	chunk index, chunk duration, chunks per episode
$L, \mathcal{B}=\{0, \dots, L-1\}$	number of ladder rungs, rung index set
$a_k^{\text{raw}}, a_k^{\text{exec}}, \iota_k$	proposed rung, executed rung, intervention bit
B_k, m, B_{crit}	buffer level, safety margin, critical buffer
$V_k(j), \sigma_k(j)$	VMAF and size (bits) of rung j at chunk k
$\varepsilon, \varepsilon_{\text{eff}}, \varepsilon_{\text{risk}}$	perceptual budget, effective/risk-mode budget
$\hat{C}_k, C_k^{\text{act}}, \underline{C}_k$	throughput predictor, realized, conformal lower bound
$\alpha, 1-\alpha$	conformal miscoverage, target coverage
$r=C_k^{\text{act}}/\hat{C}, W, \kappa$	residual ratio, residual window, predictor horizon
$d_k(j), H, \varepsilon_0$	download time of rung j , look-ahead, numerical guard
μ, β	logged-QoE smoothness and stall weights ((2))
w, w^*	QoE stall weight (sweep), mean-QoE crossover weight

3.1. Streaming architecture and control loop

We work in the standard Hypertext Transfer Protocol (HTTP) adaptive-streaming abstraction [2, 3, 16]. A video origin or content delivery network (CDN) publishes L encoded representations. We index them by $\mathcal{B} = \{0, \dots, L-1\}$; the DASH/HTTP Live Streaming (HLS) manifest carries the matching per-chunk sizes and perceptual metadata. At decision step k , the client fetches chunk k while holding a playout buffer B_k . The effective trace throughput $\hat{C}_k$ is exogenous to the controller. This is not a mere modeling convenience: on sub-6 GHz and 5G/mmWave access, throughput can change abruptly under blockage, LOS/NLOS transitions, scheduling, and mobility [5, 6].

A base policy π (heuristic or learned) maps an augmented state s_k to a proposed action $a_k^{\text{raw}} \in \mathcal{B}$. Before chunk k is downloaded, the Certified Perceptual Shield $\mathcal{S}$ applies the deterministic projection

$$(a_k^{\text{exec}}, \iota_k) = \mathcal{S}(s_k, a_k^{\text{raw}}), \quad a_k^{\text{exec}} \in \mathcal{B}, \quad \iota_k \in \{0, 1\}, \quad (1)$$

The projection returns the executed representation index a_k^{exec} and an audit bit ι_k , the latter indicating whether the projection altered the proposal. Only a_k^{exec} feeds the chunk download and buffer dynamics (4); Figure 2 traces the $a_k^{\text{raw}} \rightarrow \mathcal{S} \rightarrow a_k^{\text{exec}}$ path. Decisions advance on the chunk clock, one step per playback duration T_c (4 s in our harness) [2, 3].

The two layers serve distinct roles. Optional Lagrangian PPO training handles stall and smoothness budgets in expectation over rollouts [30, 31], yet it offers no

guarantee of per-step feasibility under conformal throughput bounds; CPS supplies this guarantee through decision-time banking and certified repair [9, 10].

3.2. State, action, reward, and nonlinear perceptual mapping

State s_k . The observation holds buffer occupancy, past throughput samples, and the previous decision. It also includes manifest lookahead: upcoming chunk sizes across the representation ladder, plus spatial/temporal complexity tags (SI/TI) [2, 3, 16]. The VMAF ladder $\{V_k(j)\}_{j \in \mathcal{B}}$ completes the observation, where $V_k(j)$ is the perceptual score for chunk k at representation j [8]. This last component is essential: without $\{V_k(j)\}$, neither a learned host nor CPS can detect the content-dependent ladder saturation that bandwidth banking exploits.

Nonlinear, content-dependent fidelity. Saturation arises because quality does not scale linearly with rate: for each chunk k , the map $j \mapsto V_k(j)$ is not affine in the discrete index j . This map also varies from chunk to chunk, because differing group-of-pictures (GOP) and texture structures yield different marginal VMAF gains between adjacent rungs, making the bitrate-to-VMAF relationship both nonlinear and content-dependent. Consequently, adjacent rungs may differ by a few VMAF points or by more than ten, and the top rungs often flatten—precisely the structure that CPS banking exploits [8]. Perceptually aware, JND-based per-title encoding aims to remove this flattening at authoring time [34], whereas CPS operates at the opposite end, exploiting whatever saturation a deployed ladder still exhibits at decision time.

Action. $a_k^{\text{raw}} \in \mathcal{B}$ is the policy’s proposed representation index for chunk k ; the environment executes only a_k^{exec} .

QoE surrogate. Keeping to RobustMPC/Pensieve reporting conventions [2, 3], we log a scalar per-chunk surrogate with fidelity $q_k \equiv V_k(a_k^{\text{exec}})$:

$$\text{QoE}_k^{(\text{rep})} = q_k - \mu \text{smooth}_k - \beta \tau_k^{\text{stall}}, \quad (2)$$

where $\text{smooth}_k = |q_k - q_{k-1}|$ is the absolute VMAF change between adjacent chunks and τ_k^{stall} is the stall duration at step k . The first chunk takes a zero smoothness term. Our shared harness fixes $\beta=4.3$ per stall second and $\mu=1.0$ per absolute VMAF-point change. The symbol α remains reserved for the conformal miscoverage throughout. The rebuffering weight is a Pensieve-style penalty, rescaled onto the VMAF fidelity axis. Session QoE sums over chunks: $\sum_{k \in \mathcal{K}_e} \text{QoE}_k^{(\text{rep})}$ (Section 5). For a learned host, optional RL training substitutes a Lagrangian CMDP objective over rollouts (Appendix A) [29, 30].

3.3. Ladder saturation

The evaluated system reports one VMAF value per representation and video (pooled per-frame scores), so shield decisions operate on a session-mean ladder that does not vary with chunk index k . Table 3 summarizes session-mean ladders for 12 publicly available reference titles (six Blender animations, six Xiph DERF clips); Figure 1 illustrates per-chunk spread on a legacy four-title subset where frame-level VMAF logs were retained. Across the twelve-title suite, 67% exhibit a non-monotone session-mean ladder and mean top-rung gain (6000 vs. 2850 kbps) is only 5.61 VMAF points—strong saturation on animation titles (e.g. Big Buck Bunny at +0.03), steeper climbs on natural content (e.g. Park Joy at +14.3). Two properties make banking worthwhile. First, many adjacent pairs differ by at most one VMAF point on saturated titles, so a single-step downgrade costs almost nothing perceptually. Second, the top rungs saturate frequently, allowing the knee rule (7) to bank bytes with no material quality loss. On the monotone per-video ladder we evaluate, highest-VMAF emergency ranking among feasible indices coincides with highest-feasible-index projection. CPS instead selects the *smallest* within-budget rung—a choice that takes effect only once saturation renders downshifts near-lossless. Chunk-level inversions appear in Figure 1 for the logged subset, but banking does not depend on them: Section 5.5 shows that the knee rule remains effective and, in fact, banks more once the shield observes the raw per-chunk ladder.

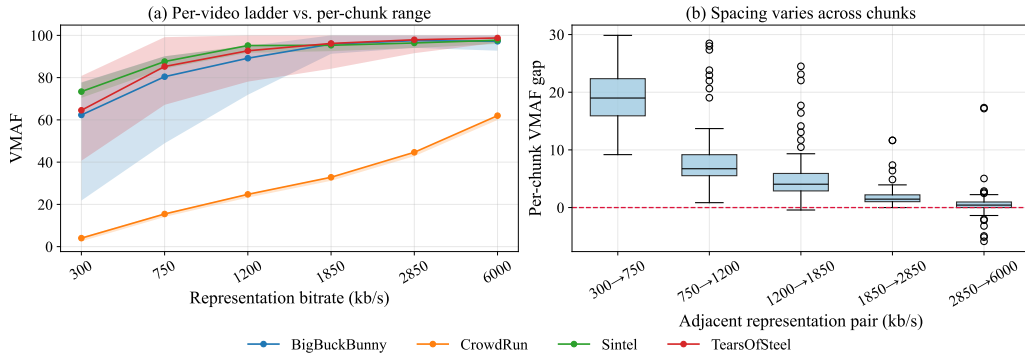


Figure 1: Representation ladder measured from the per-frame VMAF logs. (a) Per-video pooled ladder (markers) against the range spanned by individual chunks (shaded). (b) Distribution of the VMAF gap between adjacent representations across all 173 chunks; the dashed line marks a zero gap, below which the higher representation scores worse.

Table 3: Session-mean representation ladder across twelve evaluation titles.

Video	Held-out	SI	TI	Top gain	Span	Mono.
Big Buck Bunny	—	60.6	47.1	+0.03	30.6	Yes
Tears of Steel	—	41.0	31.8	+4.09	26.1	No
Sintel	✓	50.4	14.7	+1.30	22.9	Yes
Elephants Dream	—	51.8	27.0	+7.87	31.2	No
Ducks Take Off	—	94.9	16.6	+11.59	36.6	No
Cosmos Laundromat	✓	42.6	38.3	+5.23	34.6	No
Park Joy	—	119.7	34.8	+14.31	46.5	No
Old Town Cross	—	61.2	12.6	+6.85	49.5	Yes
Rush Hour	—	27.7	11.7	+4.00	29.8	No
Into Tree	—	47.4	14.4	+9.12	36.1	No
Sunflower	—	38.0	16.5	+1.42	24.9	No
Kristen & Sara	✓	85.8	7.0	+1.53	37.6	Yes

Top gain is VMAF(6000)–VMAF(2850) in kbps; Span is VMAF(6000)–VMAF(300). Mono. marks a non-decreasing session-mean ladder. Held-out titles (✓) are excluded from PPO training. SI/TI are spatial/temporal complexity indices.

3.4. Chunk-timescale dynamics

3.4.1. Throughput and latency

Let $\hat{C}_k > 0$ be the trace-derived effective throughput, in kb/s, for chunk k . Every method sees the identical throughput trace in paired evaluations. The choice of a_k^{exec} sets the manifest payload size $\sigma_k(a_k^{\text{exec}})$ in bits. Write $\tilde{C}_k = \max\{\hat{C}_k, \varepsilon_0\}$ for a small constant $\varepsilon_0 > 0$ (the same numerical guard as in (5)). The nominal download duration is $d_k = \sigma_k(a_k^{\text{exec}})/(10^3 \tilde{C}_k)$ seconds [2].

3.4.2. Buffer and stall

Let $B_k \geq 0$ be the buffered playback time when the download of chunk k begins [2, 3]. Rebuffering (stall) and the buffer update follow

$$\tau_k^{\text{stall}} = \max\{0, d_k - B_k\}, \quad (3)$$

$$B_{k+1} = \max\{0, B_k - d_k\} + T_c. \quad (4)$$

Randomness across episodes has a single source: trace and initial-condition sampling [13, 35, 36]. Given $(s_k, a_k^{\text{exec}}, \hat{C}_k)$, each transition is deterministic.

3.5. Optional Lagrangian training for RL hosts

When the base policy is learned, we may train PPO with a Lagrangian-relaxed constrained MDP (CMDP) objective that penalizes rebuffering and smoothness alongside fidelity [29–31]. The dual variables adapt as training proceeds; Appendix A details the optional buffer-deviation and Lyapunov stabilizers [37]. None of this certifies per-chunk feasibility, which finite-sample PPO with clipped objectives cannot provide [4]; CPS supplies this layer at deployment time. Co-design wraps the training environment in CPS, so the actor–critic observes the projected action distribution, whereas shield-at-eval-only policies train without it.

3.6. Certified Perceptual Shield as a hard projection

Let $V_k(j)$ be the VMAF of chunk k at rung j , $\sigma_k(j)$ its size in bits, and B_k the buffer in seconds. CPS first computes a conformal throughput lower bound $\underline{C}_k$ (kb/s), then the download time it implies,

$$d_k(j) = \min \left\{ \frac{\sigma_k(j)}{\underline{C}_k \cdot 10^3 + \varepsilon_0}, 60 \right\}. \quad (5)$$

Safety set. Indices that never exceed the proposal and fit the buffer margin m form

$$\mathcal{A}_{\text{safe},k}(a) = \{j \leq a : d_k(j) \leq \max\{B_k - m, 0.1\}\}. \quad (6)$$

VMAF-knee banking. Even when a is feasible, CPS selects the *smallest* rung within a perceptual budget ε of the proposal,

$$j_k^{\text{knee}}(a) = \min\{j \leq a : V_k(a) - V_k(j) \leq \varepsilon\}, \quad (7)$$

then projects $j_k^{\text{knee}}(a)$ onto $\mathcal{A}_{\text{safe},k}(a)$; safety always takes precedence over ε . Banking exploits *saturation*: when the top rungs carry near-equal VMAF, (7) frees bytes at no material perceptual loss and banks them as buffer.

4. Method: Certified Perceptual Shield

CPS is a deterministic, model-agnostic wrapper around any base policy that proposes $a_k^{\text{raw}} \in \mathcal{B}$. The rest of this section covers the conformal estimator, the banking and feasibility projection (Algorithm 1), a predictive extension, optional PPO co-training, and the evaluation arms.

4.1. Conformal throughput lower bound

The point predictor is the harmonic mean of the last κ observed throughputs, RobustMPC-style [2]. After each step we drop the multiplicative residual $r = C^{\text{act}}/\hat{C}$ into a rolling window of size W . With n residuals in hand, the split-conformal α -quantile takes the finite-sample rank $\lfloor \alpha(n+1) \rfloor$ (1-based). A linearly interpolated empirical quantile is excluded: it under-covers on short episodes [10, 11]. If that rank is undefined— n too small—we fall back to a fixed scale $\rho_{\text{fb}}=0.80$. The lower bound reads $\underline{C}_k = \hat{C}_k \cdot q_\alpha$, clipped to stay at or below $\hat{C}_k$. Under residual exchangeability, the construction earns $\mathbb{P}(C^{\text{act}} \geq \underline{C}_k) \geq 1 - \alpha$.

Proposition 1 (Per-chunk stall certificate). *Fix chunk k , executed index j , buffer B_k , and conformal lower bound $\underline{C}_k$ computed with miscoverage α . If $d_k(j) \leq \max\{B_k - m, 0.1\}$ via (5) and the realized throughput satisfies $C_k^{\text{act}} \geq \underline{C}_k$, then $\tau_k^{\text{stall}} = 0$. Under exchangeability of conformal residuals, $\mathbb{P}(C_k^{\text{act}} \geq \underline{C}_k) \geq 1 - \alpha$ at each step where the bound is defined.*

Proof sketch. When $C_k^{\text{act}} \geq \underline{C}_k$, the realized download time is at most $d_k(j)$, which by assumption does not exceed B_k ; hence (4) gives $\tau_k^{\text{stall}} = 0$. Coverage follows from standard split-conformal prediction with the $(n+1)$ rank correction [10, 11]. $\square$

Remark. Banking may run a lower rung than the proposal, and a lower rung can only shorten $d_k(\cdot)$. So once (6) holds, safety projection keeps the certificate intact.

Scope of the guarantee. Let us be precise about what the certificate promises and what it does not. It gives *marginal* coverage over the exchangeable residual window: $\mathbb{P}(C_k^{\text{act}} \geq \underline{C}_k) \geq 1 - \alpha$, averaged over draws. It does not give coverage *conditional* on the current buffer or throughput regime; distribution-free split conformal cannot, absent stronger assumptions [11].

If drift breaks exchangeability, coverage decays gently; the decay is bounded by a term involving the total-variation distance between calibration and test residuals [38], and a weighted-conformal variant can recover it by down-weighting stale residuals. A further subtlety arises: the online sliding window leaves residuals only *approximately* exchangeable under nonstationarity—the recent ones both calibrate the quantile and precede the predicted chunk—so the finite-sample bound is exact only on an exchangeable window. Given all this, we read Proposition 1 as the design target and check it empirically through reported coverage (Section 5), which clears $1 - \alpha$ on the primary synthetic 5G test episodes (0.919 vs. 0.90); on real broadband and at tighter α it can sit marginally below the target (Section 5.3).

Algorithm 1 Certified Perceptual Shield (compact)

1: **Input:** $a^{\text{raw}}, B_k, \{\sigma_k(j), V_k(j)\}$, conformal estimator; parameters $(\varepsilon, m, B_{\text{crit}})$
2: **Output:** (a^{exec}, ι) with $\iota = \mathbf{1}[a^{\text{exec}} \neq a^{\text{raw}}]$
3: $a \leftarrow \text{clip}(a^{\text{raw}})$; $\underline{C}_k \leftarrow$ conformal lower bound; define $d_k(\cdot)$ via (5)
4: **if** $B_k \leq B_{\text{crit}}$ **then**
5: **return** $(0, \mathbf{1}[a \neq 0])$
6: **end if**
7: compute ε_{eff} (Section 4.3); $j \leftarrow j_k^{\text{knee}}(a)$ via (7) with ε_{eff}
8: **while** $j > 0$ and $d_k(j) > \max\{B_k - m, 0.1\}$ **do**
9: $j \leftarrow j - 1$
10: **end while**
11: $a^{\text{exec}} \leftarrow \min\{a, j\}$; **return** $(a^{\text{exec}}, \mathbf{1}[a^{\text{exec}} \neq a])$

4.2. Banking and certified feasibility

Banking distinguishes CPS from prior VMAF-*ranking* shields, and the distinction is substantive rather than cosmetic. On the monotone ladders seen in practice, a ranking shield provably banks nothing.

Proposition 2 (Ranking shields collapse on monotone ladders). *Fix chunk k and suppose the VMAF ladder is nondecreasing in the rung index, $V_k(0) \leq \dots \leq V_k(L-1)$, and that download time $d_k(\cdot)$ is nondecreasing in the rung index. Let $\mathcal{F} = \{j : d_k(j) \leq B_k - m\}$ be the feasible set. Then any shield that returns $\arg \max_{j \in \mathcal{F}} V_k(j)$ returns the highest feasible index $\max \mathcal{F}$, independently of any perceptual-loss budget ε .*

Proof. Because d_k is nondecreasing, $\mathcal{F}$ is a prefix $\{0, \dots, j_{\max}\}$ with $j_{\max} = \max \mathcal{F}$. Since V_k is nondecreasing, $\arg \max_{j \in \mathcal{F}} V_k(j) = j_{\max}$; the budget ε never enters the maximization. $\square$

So retaining-VMAF ranking is just highest-feasible-index projection under another name, and it frees no bandwidth. CPS uses the *proposal's* quality instead and steps down to the smallest rung within ε of it (7). This mechanism lets CPS exploit rate–quality *saturation*: when top rungs carry near-identical VMAF, the knee lies well below $j_{\max}$, and the freed bytes bank as buffer.

Algorithm 1 realizes (7)–(6). If $B_k \leq B_{\text{crit}}$, CPS forces the lowest rung; otherwise it computes the knee under the effective budget ε_{eff} (equal to ε unless predictive mode is on; Section 4.3), then walks the index down until (5) respects the margin m . The shield never upgrades a proposal. All the while, diagnostics log banked bits, VMAF surrendered, and whether realized throughput met $\underline{C}_k$ (empirical coverage).

Evaluation arms.. RAW: run a^{raw} unshielded. SAFETY: Algorithm 1 with banking switched off ($j^{\text{knee}}=a$), leaving conformal feasibility alone. CERTIFIED: the full CPS, banking plus feasibility. Paired seeds keep the Wilcoxon and TOST tests valid across arms.

4.3. Predictive and risk-aware perceptual budget

The predictive budget widens when slack is scarce: it banks lightly when buffer headroom is ample and more aggressively when risk arises. While the buffer stays comfortable, $\varepsilon_{\text{eff}}=\varepsilon$ holds quality steady. Two triggers widen the budget: B_k drops below a risk threshold B_{risk} , or a short H -step look-ahead under a planning throughput floor (a low quantile of recent observations) foresees a dip below B_{risk} ,

$$\varepsilon_{\text{eff}} = \varepsilon + (\varepsilon_{\text{risk}} - \varepsilon) \cdot f(B_k, \text{lookahead}), \quad (8)$$

with $f \in [0, 1]$, letting the shield *pre-bank* more deeply before stalls force emergency cuts. The “improved” experiments run with $\varepsilon=1.0$, $\varepsilon_{\text{risk}}=4$, $B_{\text{risk}}=8$ s, and $H \geq 6$ at forecast quantile 0.2.

4.4. Optional learned hosts and co-design

CPS asks for no learned policy. When the host is an RL policy, we train PPO (optionally Lagrangian-shaped) with Stable-Baselines 3 [4, 39] on content-aware observations: throughput history, buffer, SI/TI, VMAF ladder. *Co-design*: wrap the training environment in CPS, and the action distribution the actor–critic sees changes. The policy can then learn to propose bolder rungs that banking will trim in safety. To gauge the effect, we compare two policies against each other under identical 5G train/test provenance: one trained without the shield (shield-at-eval-only), one co-trained with CPS. Both face the improved shield at test time.

4.5. Implementation notes

At the primary 5G operating point the defaults are $\alpha=0.10$ (coverage target 0.90), window $W=200$, $\kappa=5$, margin $m=0.5$ s, $B_{\text{crit}}=0.3$ s, and buffer cap 12 s. Table 4 collects the shield and predictive-banking defaults from the hero experiments; empirical coverage is the fraction of steps with $C^{\text{act}} \geq \underline{C}_k$. Per decision, the projection runs in $O(L+H)$ over the L ladder rungs and an H -step look-ahead, plus $O(1)$ amortized conformal-quantile upkeep on the length- W residual window. Across 20,000 rollout decisions on a commodity CPU ($L=6$, full predictive shield), per-decision wall time is 131 μs at the median and 310 μs at the 99th percentile—roughly 10^{-5} of the 4 s chunk budget. CPS runs inline, and the control loop is unaffected.

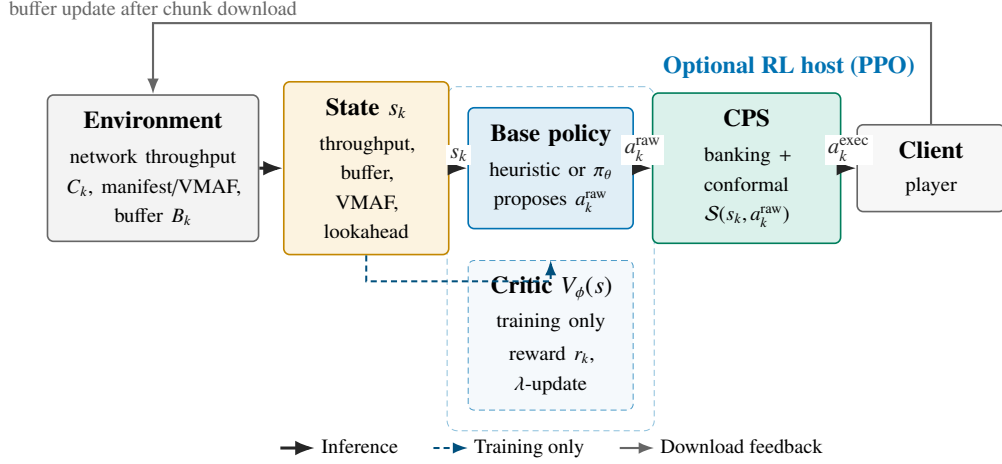


Figure 2: Control loop. The base policy proposes a_k^{raw} ; CPS banks on the VMAF knee and projects under a conformal throughput lower bound to produce a_k^{exec} . Training-only dashed paths (critic, dual variables) are optional for RL hosts.

Table 4: CPS default hyperparameters (primary 5G suite).

Parameter	Value
Perceptual budget ε	1.0 VMAF point
Conformal miscoverage α	0.10 (target coverage 0.90)
Residual window W / predictor κ	200 / 5
Buffer margin m / critical B_{crit}	0.5 s / 0.3 s
Risk budget $\varepsilon_{\text{risk}}$ / B_{risk}	4.0 / 8 s
Look-ahead horizon H / forecast quantile	≥ 6 / 0.2

5. Experimental evaluation

We organize the evaluation around six research questions:

- RQ1** *Banking vs. safety.* Does VMAF-knee banking reduce bitrate and rebuffering relative to conformal safety-only projection, with VMAF inside $\pm\varepsilon$?
- RQ2** *Coverage.* Does empirical conformal coverage meet the finite-sample target $1-\alpha$?
- RQ3** *Predictive banking.* Does a risk-aware, look-ahead budget deepen savings without leaving the perceptual envelope?

RQ4 *Co-design.* How does training with CPS in the loop change the certified operating point relative to shield-at-eval-only operation?

RQ5 *Per-chunk robustness.* Does banking remain sound and near-lossless when the shield operates on the true (non-monotone) per-chunk ladder rather than the pooled session-mean ladder?

RQ6 *Hyperparameter sensitivity.* Are the results stable across the perceptual budget ε and miscoverage α , and does coverage track the target $1-\alpha$?

5.1. Experimental setup and datasets

The testbed is a multi-video, chunk-timescale emulator ($K_c=48$ chunks/episode, $T_c=4$ s, buffer cap 12 s unless noted), driving twelve publicly available reference titles with session-mean VMAF ladders [8]: six Blender open animations (Big Buck Bunny, Tears of Steel, Sintel, Elephants Dream, Ducks Take Off, Cosmos Laundromat) and six Xiph DERF clips (Park Joy, Old Town Cross, Rush Hour, Into Tree, Sunflower, Kristen & Sara). We selected them for *ladder structure*, not catalogue breadth: saturation-rich animation (Big Buck Bunny, Tears of Steel), steep natural motion (Park Joy, Ducks), near-flat easy codecs (Into Tree, Old Town Cross), and mixed held-out titles (Sintel, Cosmos, Kristen & Sara) for generalization checks (Table 3). Nine titles appear in PPO co-design training; three are held out entirely (Sintel, Cosmos Laundromat, Kristen & Sara). The negative broadband result (Section 5.3) shows we do not select regimes that favor banking; CPS is a deterministic per-chunk projection fixed by the observed VMAF ladder and buffer state, so ladder shape—not semantic content—drives its effect. The twelve-title suite therefore spans both saturated and steep/inverted session-mean ladders (67% non-monotone).

Primary 5G suite. The primary pool is synthetic 5G traces, laced with intermittent dips and outages, held out from the co-design training seeds (204 paired episodes). We generate them synthetically because public per-chunk 5G/mmWave throughput logs fit for chunk-level ABR replay remain scarce. For realism, the generator is tuned to reported 5G/mmWave dip-and-outage dynamics [5, 15], and the broadband suite below adds a real-network counterpart. Train and test trace pools stay source-disjoint. Co-design policies train on the training pool alone and face held-out test seeds.

Broadband suite. *Real-measured* broadband traces: the FCC Measuring-Broadband-America fixed-line logs [35] and the Norway/Nor-net mobile-broadband mobility logs [36] (27 held-out test traces, source-disjoint from training). Both are

widely used public datasets in ABR research. They stand as the real-network foil to the synthetic 5G suite, and they show where banking offers limited headroom.

Hosts. Five host families cover the ABR design space: bitrate-greedy (forever at the top rung), buffer-based ABR (BBA-like), BOLA (the Lyapunov buffer-based rule of the dash.js reference player [17, 40]), robust model-predictive control (RobustMPC [2], a five-segment planner over a throughput prediction discounted by recent worst-case error), and PPO policies (content-aware “proposed” and content-blind Pensieve-style). CPS carries no host bias: one wrapper, applied at inference across all five families, tuned to none of them.

Arms and pairing. Every host rolls out RAW, SAFETY, and CERTIFIED on identical Python seeds—the same video–trace pairs. We report mean delivered bitrate, total rebuffering, stall-free rate, mean VMAF, intervention rate, and empirical conformal coverage. Significance rests on paired Wilcoxon signed-rank tests (zero-method `zsplit`) and TOST equivalence of VMAF within $\pm\epsilon=1.0$ [41], at defaults $\alpha=0.10$ and coverage target 0.90. The p -values we quote are raw and two-sided. Since several paired contrasts run at once (bitrate and rebuffering across the host families and two shield settings), we checked that every reported bitrate and rebuffering effect survives Holm–Bonferroni correction at 0.05 over the reported family; the paired-episode bootstrap backs the co-design VMAF and rebuffering contrasts on top of that. One caveat about diversity is worth stating. Each episode is one (video, trace) pair—video drawn uniformly from the twelve titles, trace from the held-out pool. Observations therefore *cluster by title*, and effective independent-content diversity is twelve, not 204.

We treat the paired tests as evidence of *within-pair* shield effects—shield against itself-off on the very same video–trace pair—and we report per-title ladder structure (Table 3) with held-out breakdown where applicable. Vanishingly small p -values (say $< 10^{-4}$) track sign-consistency of a paired difference over 204 pairs. Read them as strong evidence for a within-pair effect, not as precise magnitudes.

5.2. Model-agnostic banking on 5G (RQ1–RQ2)

Table 5 lays out absolute paired-arm metrics on synthetic 5G test traces for five host families: greedy, BBA, BOLA, RobustMPC, and a Pensieve-style negative control. Figure 3 then distills the CERTIFIED vs. SAFETY contrast across hosts and regimes. Banking behaves consistently across all four deterministic hosts on the twelve-title suite. Certified–safety bitrate reductions read -4.5% , -4.1% , -4.1% , and -3.8% , with rebuffering cuts of -9.3% , -8.9% , -8.9% , and -9.2% (Table 5; greedy $p=8.5 \times 10^{-4}$). Steep and non-saturated ladders (Table 3) keep absolute savings modest—the saturation-dependent regime the mechanism predicts—yet

every host stays perceptually near-lossless. Mean VMAF differences are perceptually negligible, at -0.01 , 0.17 , 0.18 , and 0.02 points across greedy, BBA, BOLA, and RobustMPC, with every value inside $\pm\epsilon$ by TOST. The two shield layers split the work. Against RAW, certified greedy cuts rebuffering by -55.5% (BBA: -51.0%). Conformal safety eliminates the catastrophic stalls; banking adds the rest of the headroom. Coverage follows a distinct pattern on the synthetic 5G pool. Empirical coverage there is 0.919 (greedy) and 0.919 (BBA) against a 0.90 target, consistent with the $(n+1)$ conformal calibration at these episode lengths. Because coverage depends on the throughput trace and the conformal calibrator rather than the base policy, it reads identically (0.919) across every host and shield variant on the 5G test set (Figure 4). Proposition 1 is a *per-chunk* probabilistic certificate, not a per-episode stall-free one: at coverage 0.919 , about 8% of chunks can watch realized throughput slip below $\underline{C}_k$, and the forced downshift at B_{crit} plus the first-chunk cold start explain the residual rebuffering in the certified arm. Pensieve, the negative control, degrades to near-floor bitrates with near-zero stalls, so CPS rarely fires—banking cannot rescue a policy that never asks for saturated rungs.

That high intervention rate under greedy (Table 5) is fixed *by construction*; do not mistake it for the shield dominating an informed base policy. Greedy proposes the top rung every time, so nearly every banking downshift or feasibility repair shows up as an intervention. The rate gauges how far a host sits from the certified operating point, not how much “credit” the shield deserves. That is precisely why we run the model-agnostic contrast across hosts spanning the spectrum.

Certified intervention decreases across all hosts: 70.4% for always-top greedy, 61.1% for BOLA, a mere 31.1% for RobustMPC—which already plans conservatively against a discounted throughput estimate—and near zero for the floor-hugging Pensieve-style control. The sharper the base policy, the lighter the shield’s load. Yet coverage holds at 0.919 throughout, and the certified metrics lean more and more on the base policy’s own calls, nudged only where banking headroom survives.

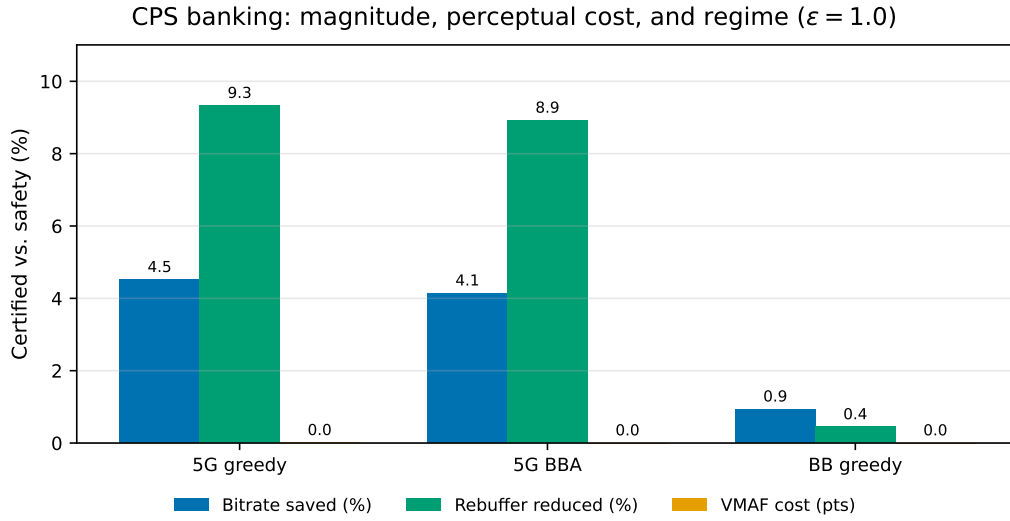


Figure 3: CPS banking overview (certified vs. safety, 204 paired episodes, $\varepsilon=1.0$). For two synthetic 5G hosts and a broadband host, bars show percent bitrate saved, percent rebuffering reduced, and mean VMAF cost (points). Banking is significant and perceptually near-lossless on 5G (magnitude scales with ladder saturation) but collapses on broadband, where top rungs are rarely jointly feasible and saturated.

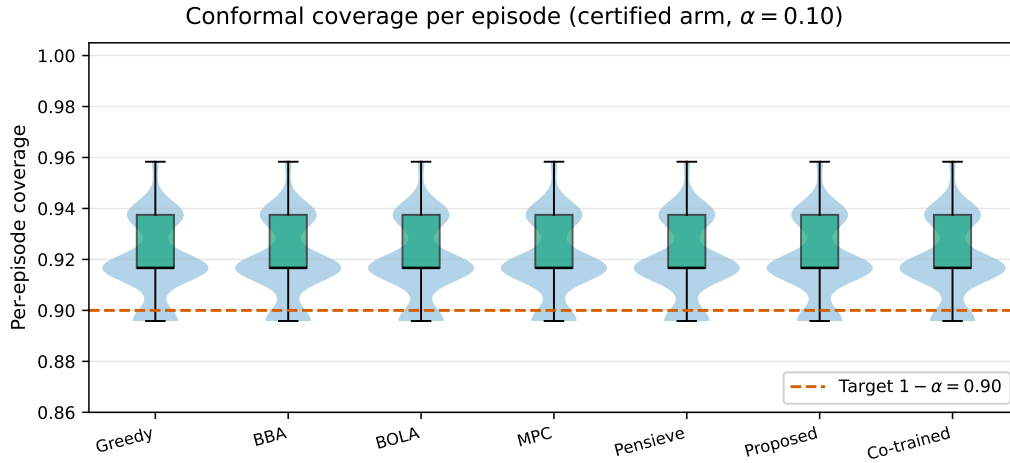


Figure 4: Per-episode conformal coverage of the certified arm on the synthetic 5G test pool ($\alpha=0.10$, target 0.90, dashed). Coverage depends only on the throughput trace and calibrator, so the distribution is host-invariant by construction on this pool; empirical coverage tracks the finite-sample target closely (RQ2). At tighter miscoverage ($\alpha=0.05$, Table 9) or on real broadband traces (Section 5.3), coverage can sit marginally below the nominal $1-\alpha$ at $n=204$ episodes.

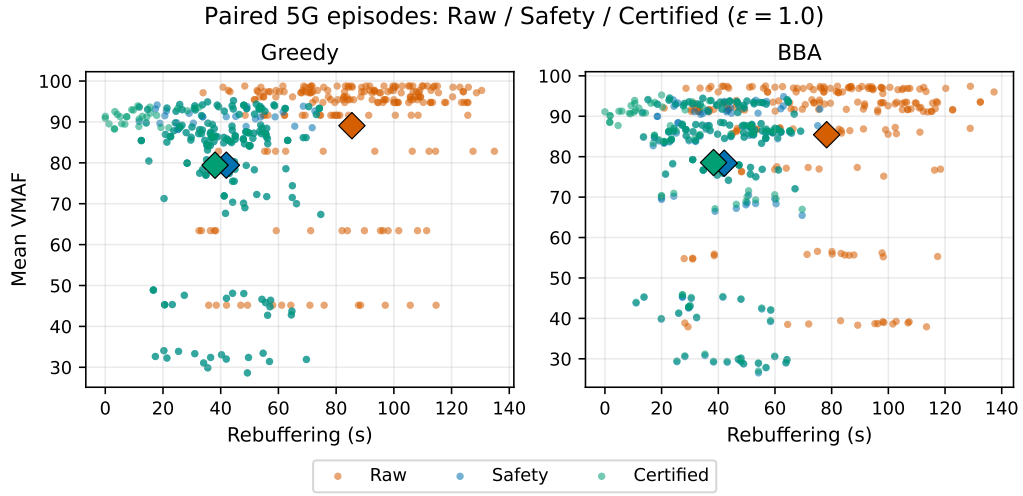


Figure 5: Per-episode operating points on synthetic 5G (paired seeds). Diamonds mark arm means. Safety removes catastrophic stalls relative to Raw; certified banking further reduces rebuffering at nearly unchanged mean VMAF.

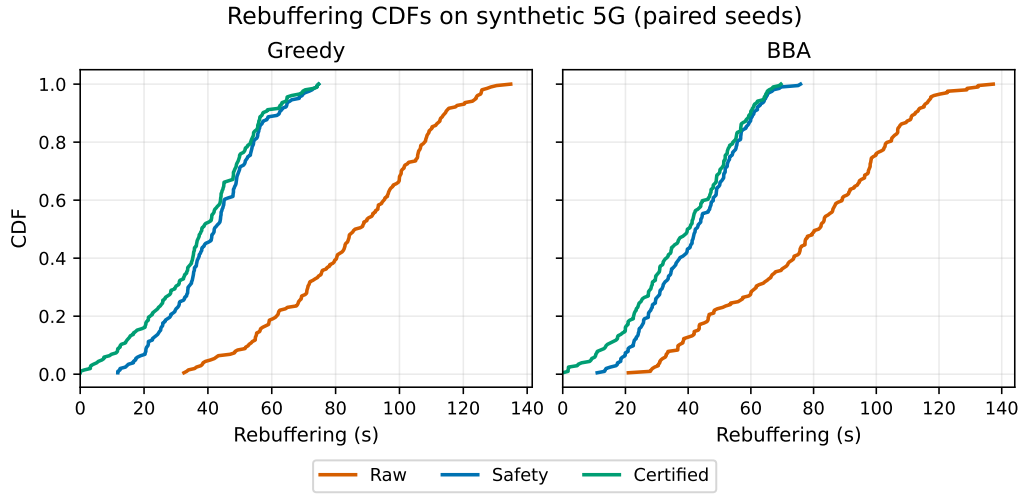


Figure 6: Empirical CDFs of episode rebuffering for greedy and BBA hosts. The certified arm stochastically dominates safety and raw on the stall axis.

Table 5: Paired arms on synthetic 5G test traces (204 episodes, $\varepsilon=1.0$, $\alpha=0.10$). Reb. is total rebuffering; Cover. is empirical conformal coverage; Interv. is shield intervention rate.

Host	Arm	Reb.	VMAF	Bitrate	Cover.	Interv.
Greedy	RAW	85.5 s	89.1	6000	—	0.0
	SAFETY	41.9 s	79.4	3131	0.919	65.8%
	CERTIFIED	38.0 s	79.4	2990	0.919	70.4%
BBA	RAW	78.2 s	85.4	4662	—	0.0
	SAFETY	42.1 s	78.4	2809	0.919	58.1%
	CERTIFIED	38.3 s	78.5	2692	0.919	66.1%
BOLA	RAW	77.6 s	83.2	4627	—	0.0
	SAFETY	42.4 s	77.5	2782	0.919	54.2%
	CERTIFIED	38.7 s	77.7	2667	0.919	61.1%
RobustMPC	RAW	71.3 s	77.4	3175	—	0.0
	SAFETY	40.0 s	74.9	2377	0.919	28.3%
	CERTIFIED	36.3 s	74.9	2287	0.919	31.1%
Pensieve	RAW	0.1 s	72.2	722	—	0.0
	SAFETY	0.1 s	70.8	689	0.919	7.3%
	CERTIFIED	0.1 s	70.8	689	0.919	7.4%

Certified–safety Wilcoxon: greedy bitrate $p=8.5\times 10^{-4}$, rebuffering $p=8.5\times 10^{-4}$; BBA bitrate $p=4.2\times 10^{-16}$, rebuffering $p=2.5\times 10^{-4}$. Improved predictive banking defaults in Table 4.

5.3. Predictive banking and broadband boundary (RQ3)

With risk-aware $\varepsilon(B)$ and dip forecasting enabled, the certified–safety bitrate reduction climbs to -14.6% (greedy) and -12.5% (BBA), and rebuffering falls -12.6% and -13.0% . Mean VMAF changes remain negligible (-0.23 , -0.02).

The broadband suite marks the edge of these gains. On source-disjoint *real-measured* broadband traces (FCC MBA and Norway/Nor-net [35, 36]), the greedy certified–safety bitrate change dwindles to -0.9% : top rungs are seldom feasible and saturated at once, so banking has almost nowhere to go. Even so, safety vs. raw still cuts rebuffering by -98.7% . Under broadband stress the conformal feasibility layer—not banking—accounts for most of the improvement. On the same real-measured traces at $\alpha=0.10$, empirical conformal coverage is 0.890 (below the 0.90 target at $n=204$ episodes), whereas the synthetic 5G pool reaches 0.919 (Table 5, Figure 4). The “BB greedy” group in Figure 3 makes the regime split explicit: banking savings are largest on synthetic 5G. This split is by design, not a flaw: CPS banking pays off precisely when the ladder knee sits within reach, and broadband

places it out of reach.

5.4. Learned hosts and co-design (RQ4)

Two PPO settings warrant separate reporting. The *primary* host is a content-aware policy trained on source-disjoint broadband traces and evaluated, untouched, on 5G—the model-agnostic deployment case. The *co-design* pair are two policies trained on matched 5G traces—one shield-at-eval-only, one co-trained with CPS in the loop—set side by side on their certified arms (Table 6 and Figure 7). On the primary broadband-trained policy, certified banking vs. safety returns -35.3% bitrate and -79.3% rebuffering ($p=6.2\times 10^{-18}$), at VMAF $\Delta = -0.33$ (TOST within $\pm\epsilon$). The effect exceeds greedy or BBA because the learned policy still proposes high rungs often on the twelve-title suite, leaving more saturation headroom for CPS. Under the improved shield, banking reaches $-21.9\%/-29.6\%$.

Co-design. Both policies train on matched 5G traces (source-disjoint test). Under the improved shield at eval, the co-trained policy’s certified arm hits VMAF 75.73 and bitrate 1463 kb/s, against 72.80/801 for shield-at-eval-only operation. The Δ row in Table 6 is the difference of those certified-arm means ($+2.93$ VMAF, $+0.55$ s rebuffering), and it matches the raw rows above ($75.73 - 72.80, 0.86 - 0.31$). Weigh the fidelity gain against its stall cost through $\text{QoE}(w) = \overline{\text{VMAF}} - w \cdot \text{rebuffer}(s)$. The mean-QoE crossover is $w^*=5.31$ VMAF points per second (ratio of the Δ row entries): co-design favors fidelity-oriented weights $w < w^*$; shield-at-eval-only prevails where stall aversion is stronger. Paired Wilcoxon tests and episode-bootstrap intervals are reported as 5.0×10^{-5} (VMAF) and 5.0×10^{-5} (rebuffering). The bootstrap CI is $[2.61, 3.25]$, with stall cost $[0.29, 0.85]$ s and crossover $[3.40, 10.27]$ (median 5.37; positive VMAF in 100.0% of resamples). Co-design remains a supporting result under a single training seed (Section 6).

Table 6: Co-design on 5G test (certified arm, improved shield at eval).

Policy	VMAF	Reb. (s)	Bitrate	Cover.
Eval-only shield	72.80	0.31	801	0.919
Co-trained (CPS)	75.73	0.86	1463	0.919
Δ (co – eval)	+2.93	+0.55	—	—

The Δ row equals the certified-arm difference in the rows above. Paired Wilcoxon: VMAF $p=5.0\times 10^{-5}$, rebuffering $p=5.0\times 10^{-5}$; bootstrap ΔVMAF CI $[2.61, 3.25]$ over 204 episodes (requires synced `episodes.csv`).

5.5. Per-chunk (non-monotone) ladder ablation (RQ5)

The headline runs apply the knee rule to the pooled session-mean ladder, a monotone one. We therefore test whether banking survives on the *true per-chunk*

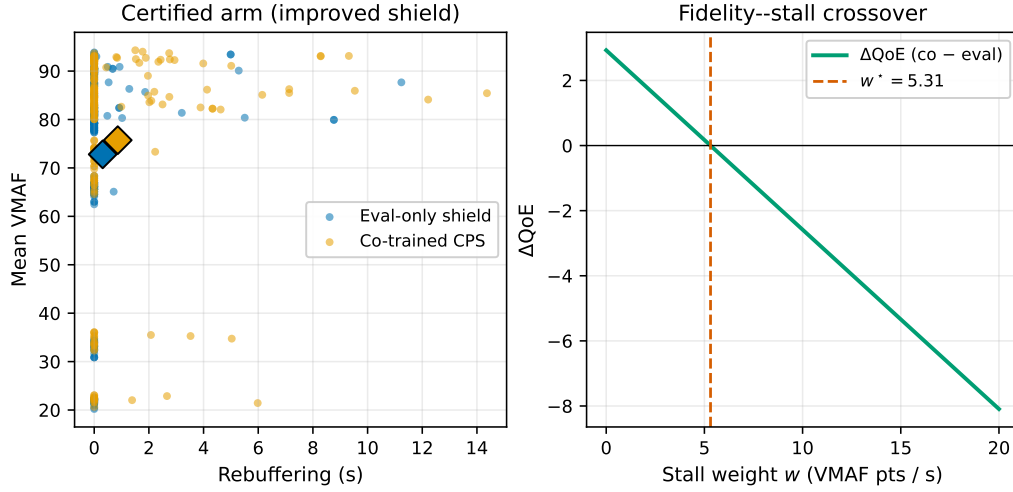


Figure 7: Co-design on 5G test (certified arm, improved shield). Left: per-episode VMAF–rebuffer points (diamonds = means). Right: mean $\Delta QoE(w)$ versus stall weight; the dashed line marks the crossover $w^*=5.31$.

ladders, which often turn non-monotone (26.5% of the 113 chunks hold at least one inverted adjacent pair; Table 3). To settle it, we replay the deterministic banking stage on the per-chunk logs, pitting a shield that sees only the pooled ladder against one that sees the per-chunk ladder (Table 7, Figure 8). Banking alone depends on VMAF; the feasibility projection ignores it and stays identical in both cases, which isolates the effect of the pooling simplification. Three findings hold at the default $\varepsilon=1.0$. First, the knee rule stays well defined on inverted ladders and banks *more*, not less—mean saved bitrate is 45.8% per-chunk against 44.6% pooled, since a lower rung that inverts above the proposal clears the budget outright. Second, a per-chunk shield holds realized per-chunk perceptual cost inside budget *by construction* (mean 0.32 VMAF points, never above ε). Third, pooling is not free. Judged against the true per-chunk VMAF, the pooled-ladder shield overruns the budget on 11.5% of chunks (mean cost 0.33 points). Put the three together and the pooled-ladder headline numbers look, if anything, *conservative* about how much banking wins. Per-chunk operation is sound, wiping out the residual budget violations for the price of a per-chunk VMAF feed, and it bridges the content-diversity motivation with the pooled evaluation.

5.6. Sensitivity to ε and α (RQ6)

The headline runs fix the pair at $\varepsilon=1.0$ and $\alpha=0.10$. Tables 8–9 and Figure 9 replay the same greedy host on the synthetic 5G test pool at 204 paired episodes

Table 7: Banking-stage ablation on real per-chunk VMAF ladders (greedy proposal, 113 chunks). A per-chunk shield banks slightly more bytes than the pooled shield while keeping realized per-chunk cost $\leq \varepsilon$ by construction; the pooled shield incurs a small per-chunk budget-violation rate. Feasibility is VMAF-independent and unchanged.

ε	Mean knee rung		Bitrate saved (%)		Per-chunk cost (pts)	Budget viol. pooled (%)
	pooled	per-chunk	pooled	per-chunk	per-chunk	
0.5	4.58	4.34	22.3	28.1	0.02	8.8
1.0	4.15	3.94	44.6	45.8	0.32	11.5
2.0	3.60	3.39	58.2	59.7	0.98	18.6
4.0	2.93	2.88	68.7	68.7	1.89	7.1

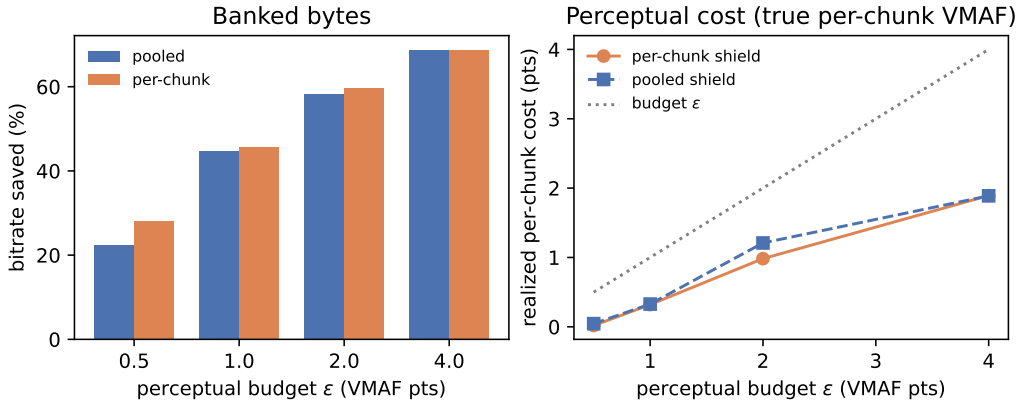


Figure 8: Per-chunk vs. pooled ladder banking across the perceptual budget ε . Left: banked bitrate is comparable (slightly larger per-chunk). Right: realized per-chunk perceptual cost stays under ε for the per-chunk shield; the pooled shield tracks it closely on average but violates the budget on a minority of chunks.

(twelve-title roster); regenerate via `ablation_eps_alpha.py` when the roster or episode count changes. We first consider the perceptual budget. Sweeping $\varepsilon \in \{0.5, \dots, 4.0\}$ (at $\alpha=0.10$) trades banking for fidelity, and it does so cleanly. Bitrate saved versus safety climbs from 2.0% to 24.6%, while the mean VMAF change stays trifling (at most 0.48 points) and never leaves the matching $\pm\varepsilon$ envelope. Empirical conformal coverage remains invariant with ε —it answers to the throughput trace and calibrator alone, not the banking budget, which shows banking and the feasibility certificate run on separate tracks by design. We next consider miscoverage. Sweeping $\alpha \in \{0.05, 0.10, 0.20\}$ (at $\varepsilon=1.0$) on the same synthetic 5G pool drags empirical coverage in step with the finite-sample target

Table 8: Sensitivity to the perceptual budget ε (greedy host, 5G test, $\alpha=0.10$, certified vs. safety). Banking scales smoothly with ε ; the mean VMAF change stays negligible and coverage is invariant to ε .

ε (pts)	Bitrate saved (%)	Rebuffer Δ (%)	VMAF Δ (pts)	Coverage
0.5	2.0	-3.5	+0.02	0.919
1.0	4.4	-8.2	-0.00	0.919
2.0	14.4	-30.8	-0.17	0.919
4.0	24.6	-46.1	-0.48	0.919

Table 9: Sensitivity to the conformal miscoverage α (greedy host, 5G test, $\varepsilon=1.0$). Empirical coverage tracks the finite-sample target $1-\alpha$, and the intervention rate rises as α tightens.

α	Target $1-\alpha$	Coverage	Interv. (%)	Bitrate saved (%)
0.05	0.95	0.945	73.0	4.5
0.10	0.90	0.919	71.0	4.4
0.20	0.80	0.854	35.2	6.2

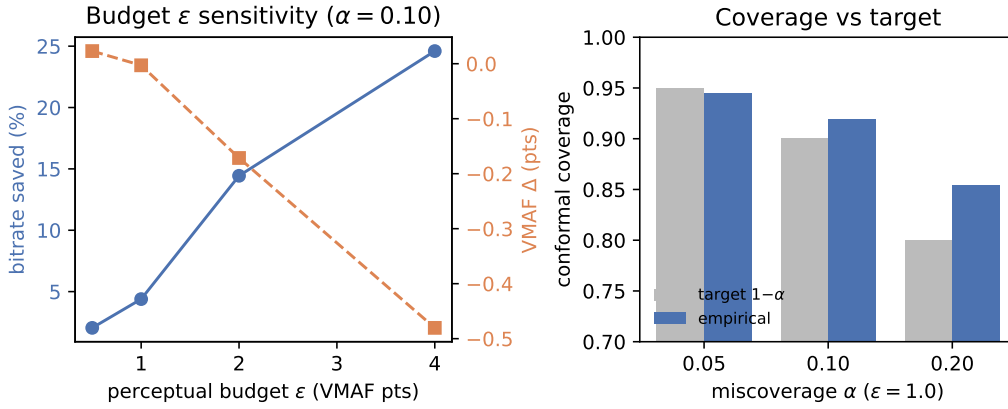


Figure 9: Hyperparameter sensitivity. Left: banked bitrate rises and the mean VMAF change stays near zero as the budget ε grows ($\alpha=0.10$). Right: empirical conformal coverage tracks the target $1-\alpha$ across α ($\varepsilon=1.0$).

$1-\alpha$ (coverage 0.945, 0.919, 0.854 for targets 0.95, 0.90, 0.80): Tighten α and both coverage and intervention rate rise; loosen it and both ease off, as the $(n+1)$ construction foretells. At $\alpha=0.05$ the mean is 0.945, marginally below the 0.95 target at $n=204$ —the same finite-sample slack seen on real broadband traces (Section 5.3). So the default operating point lies on a smooth, well-behaved frontier rather than a narrowly tuned optimum.

Reproducibility. Artifacts include evaluation scripts, exported episode CSVs,

`make_cps_paper_assets.py` for regenerating Tables 5–6 and Figures 3–7, and `ablation_eps_alpha.py` for the ε/α sensitivity study (Tables 8–9, Figure 9). The per-chunk ablation (Table 7, Figure 8) is regenerated by `ablation_perchunk_ladder.py`, which replays the banking rule on the raw per-chunk VMAF logs with no learned components.

6. Discussion and limitations

Scope. CPS is an auditable, model-agnostic mechanism that (i) banks bandwidth on VMAF-saturated ladder knees inside a stated perceptual budget and (ii) enforces download feasibility under a conformal throughput lower bound with target coverage $1-\alpha$. Its silences are deliberate. It never claims dominance over every ABR controller, nor a pathwise worst-case guarantee beyond the conformal assumption of residual exchangeability. The payoff also depends on the regime: banking gains gather where high rungs are often feasible (moderate-buffer synthetic 5G), whereas under harsh broadband stress the safety projection dominates and banking headroom thins.

Local ladder inversions. Table 3 shows that, chunk by chunk, a sizable share of adjacent rungs in titles like Sintel and Big Buck Bunny run near-flat or even *inverted*—a higher bitrate scoring no better, sometimes worse, in VMAF. CPS handles all of this without a special case. Structurally, the evaluated shield runs on the monotone session-mean ladder, so Proposition 2 applies outright; and even on a fully per-chunk, non-monotone ladder both stages stay well defined. Each stage, moreover, handles inversions on its own terms. The feasibility projection (6) watches download time, never VMAF, so inversions leave safety untouched. The knee rule (7) picks the smallest rung within ε of the proposal’s VMAF; when an inversion lets a lower rung score at least as high, that rung clears the budget on its own, and banking captures the extra headroom instead of stumbling on it. Local inversions, then, can only widen the banking opportunity, and the algorithm clears them by construction. The per-chunk ablation in Section 5.5 bears this out: banked bytes match, and realized per-chunk perceptual cost stays inside budget. Per-chunk ladders buy accuracy, not correctness.

Threats to validity.

1. The broadband suite replays *real-measured* FCC and Norway/Nor-net traces [35, 36], yet the 5G suite is synthetic, and either way the chunk-level emulator strips out transport-stack dynamics and CDN caching [5]. Live-player validation waits for future work.

2. Twelve publicly available titles and finite trace pools still cap catalogue diversity [36]; three are held out from PPO training but recur in CPS evaluation.
3. QoE here is a VMAF-based surrogate. Subjective rankings of stall against fidelity may part from it [7].
4. Conformal coverage presumes exchangeable residuals. Sharp nonstationarity can erode the guarantee, though the online window adapts locally and the coverage loss stays bounded by the residual distribution shift [38]. At $n=204$ episodes, empirical coverage can sit marginally below the nominal $1-\alpha$ target on real broadband traces (0.890 at $\alpha=0.10$) or at tight miscoverage on synthetic 5G (0.945 vs. 0.95 at $\alpha=0.05$; Table 9); headline RQ2 reports the default 5G operating point (0.919 vs. 0.90).
5. Every arm—greedy, BBA, the PPO hosts—runs locally under the same finite-sample $(n+1)$ conformal correction. Co-design magnitudes hinge on the checkpoint, so we report the best-validation model of each policy. The Δ rows of Table 6 come from certified-arm means, while the paired Wilcoxon test and bootstrap intervals use per-episode exports over the same 204 pairs, with both co-design policies re-evaluated together. One point deserves weight: the co-design contrast uses a *single* training seed per policy, so the 95% intervals speak to evaluation variance, not training-seed variance. Establishing training-seed variance for w^* would require retraining many seeds, which we leave to future work. Hence co-design stands as a supporting result, not a headline claim.
6. Banking savings *scale with ladder saturation* (Table 3); the twelve-title mix deliberately includes steep and non-saturated ladders, so greedy/BBA effect sizes are conservative relative to a saturation-rich subset alone.
7. The Pensieve-style no-op reveals that host aggressiveness sets the banking opportunity, so results should not be carried over to arbitrary policies without the paired-arm protocol.
8. The TOST margin equals the configured banking budget ε —a design parameter, not a preregistered clinical equivalence margin [41].

Future work. Several directions remain open: packet-level or field 5G evaluation; applying CPS to RobustMPC and Comyco hosts; CVaR/tail-aware objectives [42]; telemetry-driven adaptation of ε and α ; and preregistered non-inferiority studies with a fixed perceptual margin. Real-player deployment is the direction we care about most. Every result here comes from a chunk-level emulator that strips out transport-stack dynamics, so dropping CPS into a production player like `dash.js` [40] raises concrete challenges the simulation never sees; three loom

largest. First, throughput has to be read off noisy, bursty HTTP/TCP (or QUIC) samples whose exchangeability—the very assumption behind the conformal bound—can break under slow-start, congestion-control transients, and CDN re-routing; the calibration residuals would then call for periodic re-estimation, or a sliding, weighted-conformal variant that provably bounds coverage loss under drift [38]. Second, the feasibility check has to square the player’s own buffer-occupancy signal, segment-level and delayed, with the certificate’s buffer-margin accounting. Third, the per-decision budget tightens from our $\sim 100\mu\text{s}$ microbenchmark into a soft real-time constraint running beside the player’s ABR loop. Showing that the finite-sample coverage guarantee withstands these real-world perturbations—and measuring whatever coverage slack they open up—is the obvious next step.

7. Conclusion

We present the Certified Perceptual Shield (CPS) for client-side ABR, comprising three components: VMAF-knee bandwidth banking, an online conformal throughput lower bound with finite-sample coverage, and certified buffer feasibility. Across 204 paired synthetic 5G episodes over twelve ladder-diverse titles, the picture is consistent: banking against conformal safety-only cuts both delivered bitrate and rebuffering across every deterministic host ($\approx -4.5\%/ -9.3\%$ for greedy through $-3.8\%/ -9.2\%$ for RobustMPC), with VMAF inside ± 1.0 and empirical coverage at 0.919 against a 0.90 target. Predictive banking further improves the margin (-14.6% greedy bitrate); under broadband stress, safety alone predominates once the ladder knee lies beyond the observable operating range. As a supporting result, applying co-design—training PPO with CPS in the loop—shifts the certified operating point toward higher fidelity ($+2.93$ VMAF at $+0.55$ s rebuffering; Table 6), with crossover $w^*=5.31$ VMAF points per second of stall. CPS integrates with existing ABR controllers as a lightweight wrapper—no retraining required—and extends naturally to learned policies.

Appendix A. Lagrangian CMDP training for RL hosts

An unconstrained MDP chases $\mathbb{E}[\sum_k \gamma^k r_k]$ and nothing else—no explicit budget on stalls or smoothness. So when the base policy is learned, we may recast streaming as a constrained MDP (CMDP) that maximizes expected fidelity under budgets on trajectory-level auxiliary costs [29]. Set $c_k^{\text{stall}} \equiv \tau_k^{\text{stall}}$ and $c_k^{\text{sm}} \equiv \text{smooth}_k$.

The undiscounted episodic rebuffering ratio and mean smoothness cost read

$$\text{RB}(e) = \frac{1}{T_e^{\text{play}}} \sum_{k \in \mathcal{K}_e} c_k^{\text{stall}}, \quad \text{SM}(e) = \frac{1}{K_e} \sum_{k \in \mathcal{K}_e} c_k^{\text{sm}}, \quad (\text{A.1})$$

with targets $\delta_{\text{rf}}=0.05$ and $\delta_{\text{sm}}=3.5$ [2]. Per-step normalized costs $\widetilde{\text{RB}}_k$ and $\widetilde{\text{SM}}_k$ yield the Lagrangian-relaxed PPO reward

$$r_k^{\text{train}} = \frac{1}{100} (q_k - \lambda_r \widetilde{\text{RB}}_k - \lambda_s \widetilde{\text{SM}}_k - w_b \phi(B_k) - w_\ell \psi_k^{\text{Lya}}), \quad (\text{A.2})$$

where projected ascent on the empirical constraint gaps updates the dual variables (λ_r, λ_s) [30]. Training may also subtract buffer-deviation and Lyapunov drift penalties [37]; these steady the optimization but certify nothing. The gap is real: finite-sample PPO does not certify per-chunk feasibility [4, 31], and CPS hands over that layer at deployment time.

The implementation rides Stable-Baselines 3 PPO [4, 39]: two Tanh MLPs (width 256), GAE ($\gamma=0.99$, $\lambda_{\text{GAE}}=0.95$), dual step sizes $\eta_r=0.012$ and $\eta_s=0.003$, and projected ranges $\lambda_r \in [4.3, 40]$, $\lambda_s \in [0.3, 2.5]$ after a 60-episode warm-up. As in the main text, co-design wraps the training environment in CPS; shield-at-eval-only policies train without it.

Data availability

The reproducibility artifacts include the code, evaluation scripts, and exported CSV files underlying Tables 5–6 and Figures 3–7. The Certified Perceptual Shield implementation and its parameters are documented with the code.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Saeed Zarbi: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization.
Leili Farzinvash: Supervision, Writing – review & editing. **Pedram Salehpour:** Conceptualization, Supervision, Project administration, Writing – review & editing.

References

- [1] T.-Y. Huang, R. Johari, N. McKeown, M. Trunnell, M. Watson, A buffer-based approach to rate adaptation: Evidence from a large video streaming service, in: Proc. ACM SIGCOMM, 2014, pp. 187–198.
- [2] X. Yin, A. Jindal, V. Sekar, B. Sinopoli, A control-theoretic approach for dynamic adaptive video streaming over HTTP, in: Proc. ACM SIGCOMM, 2015, pp. 325–338.
- [3] H. Mao, M. Alizadeh, I. Menache, S. Kandula, Neural adaptive video streaming with pensieve, in: Proc. ACM SIGCOMM, 2017, pp. 197–210.
- [4] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, O. Klimov, Proximal policy optimization algorithms, arXiv preprint arXiv:1707.06347 (2017). arXiv:1707.06347.
- [5] A. Narayanan, X. Zhang, R. Zhu, A. Hassan, S. Jin, X. Zhu, X. Zhang, D. Rybkin, Z. Yang, Z. M. Mao, F. Qian, Z.-L. Zhang, A variegated look at 5G in the wild: Performance, power, and QoE implications, in: Proc. ACM SIGCOMM, 2021, pp. 610–625. doi:10.1145/3452296.3472923.
- [6] Cisco Systems, Cisco annual internet report (2018–2023), <https://www.cisco.com/c/en/us/solutions/executive-perspectives/annual-internet-report/index.html>, accessed: 2025 (2023).
- [7] M. Seufert, S. Egger, M. Slanina, T. Zinner, T. Hoßfeld, P. Tran-Gia, A survey on quality of experience of HTTP adaptive streaming, IEEE Communications Surveys & Tutorials 17 (1) (2015) 469–492. doi:10.1109/COMST.2014.2360940.
- [8] Z. Li, A. Aaron, I. Katsavounidis, A. Moorthy, M. Manohara, Toward a practical perceptual video quality metric, Netflix Technology Blog, <https://netflixtechblog.com/toward-a-practical-perceptual-video-quality-metric-653f208b9652> (2016).
- [9] M. Alshiekh, R. Bloem, R. Ehlers, B. Könighofer, S. Niekum, U. Topcu, Safe reinforcement learning via shielding, in: Proc. AAAI Conf. Artificial Intelligence, Vol. 32, 2018, pp. 2669–2678.

- [10] V. Vovk, A. Gammerman, G. Shafer, *Algorithmic Learning in a Random World*, Springer, 2005.
- [11] A. N. Angelopoulos, S. Bates, Conformal prediction: A gentle introduction, *Foundations and Trends in Machine Learning* 16 (4) (2023) 494–591. doi : 10.1561/2200000101.
- [12] Y. Sun, X. Yin, J. Jiang, V. Sekar, F. Lin, N. Wang, T. Liu, B. Sinopoli, CS2P: Improving video bitrate selection and adaptation with data-driven throughput prediction, in: *Proc. ACM SIGCOMM*, 2016, pp. 272–285.
- [13] A. Bentaleb, A. Taani, A. C. Begen, R. Zimmermann, S. Harous, A survey on bitrate adaptation schemes for streaming media over HTTP, *IEEE Communications Surveys & Tutorials* 21 (1) (2018) 562–585.
- [14] L. Peroni, S. Gorinsky, An end-to-end pipeline perspective on video streaming in best-effort networks: A survey and tutorial, *ACM Computing Surveys* 57 (12) (2025). doi : 10.1145/3742472.
- [15] R. A. K. Fezeu, E. Ramadan, W. Ye, B. Minneci, J. Xie, A. Narayanan, Z.-L. Zhang, et al., Unveiling the 5G mid-band landscape: From network deployment to performance and application QoE, in: *Proc. ACM SIGCOMM*, 2024. doi : 10.1145/3651890.3672269.
- [16] I. Sodagar, The MPEG-DASH standard for multimedia streaming over the internet, *IEEE MultiMedia* 18 (4) (2011) 62–67. doi : 10.1109/MMUL.2011.71.
- [17] K. Spiteri, R. Urgaonkar, R. K. Sitaraman, BOLA: Near-optimal bitrate adaptation for online videos, *IEEE/ACM Transactions on Networking* 28 (4) (2020) 1698–1711.
- [18] T. Chen, Y. Lin, N. Christianson, Z. Akhtar, S. Dharmaji, M. Hajiesmaili, A. Wierman, R. K. Sitaraman, SODA: An adaptive bitrate controller for consistent high-quality video streaming, in: *Proc. ACM SIGCOMM*, 2024. doi : 10.1145/3651890.3672260.
- [19] A. Mehrabi, M. Siekkinen, A. Ylä-Jääski, Edge computing assisted adaptive mobile video streaming, *IEEE Transactions on Mobile Computing* 18 (4) (2019) 787–800. doi : 10.1109/TMC.2018.2850026.

- [20] Z. Akhtar, Y. S. Nam, R. Govindan, S. Rao, J. Chen, E. Katz-Bassett, B. Ribeiro, Oboe: Auto-tuning video ABR algorithms to network conditions, in: Proc. ACM SIGCOMM, 2018, pp. 44–58.
- [21] M. Gadaleta, F. Chiariotti, M. Rossi, A. Zanella, D-DASH: A deep Q-learning framework for DASH video streaming, IEEE Transactions on Cognitive Communications and Networking 3 (4) (2017) 703–718. doi:10.1109/TCCN.2017.2755007.
- [22] M. Naresh, P. Saxena, M. Gupta, PPO-ABR: Proximal policy optimization based deep reinforcement learning for adaptive bitrate streaming (2023). arXiv:2305.08114.
- [23] R. S. Sutton, A. G. Barto, Reinforcement Learning: An Introduction, 2nd Edition, MIT Press, 2018.
- [24] T. Huang, C. Zhou, R.-X. Zhang, C. Wu, X. Yao, L. Sun, Comyco: Quality-aware adaptive video streaming via imitation learning, in: Proc. ACM International Conference on Multimedia (MM), 2019, pp. 429–437. doi:10.1145/3343031.3351014.
- [25] H. Zhang, A. Zhou, Y. Hu, C. Li, G. Wang, X. Zhang, H. Ma, L. Wu, A. Chen, C. Wu, Loki: Improving long tail performance of learning-based real-time video adaptation by fusing rule-based models, in: Proc. ACM MobiCom, 2021, pp. 775–788. doi:10.1145/3447993.3483259.
- [26] F. Y. Yan, H. Ayers, C. Zhu, S. Fouladi, J. Hong, K. Zhang, P. Levis, K. Winstein, Learning in situ: A randomized experiment in video streaming, in: Proc. USENIX NSDI, 2020, pp. 495–511.
- [27] X. Hu, A. Ghosh, X. Liu, Z.-L. Zhang, N. Shroff, COREL: Constrained reinforcement learning for video streaming ABR algorithm design over mmWave 5G, in: Proc. IEEE International Workshop on Communications Quality and Reliability (CQR), 2023, pp. 1–6. doi:10.1109/CQR59928.2023.10317803.
- [28] S. Gu, L. Yang, Y. Du, G. Chen, F. Walter, J. Wang, A. Knoll, A review of safe reinforcement learning: Methods, theories, and applications, IEEE Transactions on Pattern Analysis and Machine Intelligence 46 (12) (2024) 11216–11235. doi:10.1109/TPAMI.2024.3457538.

- [29] E. Altman, *Constrained Markov Decision Processes*, CRC Press, 1999.
- [30] S. Paternain, L. F. O. Chamon, M. Calvo-Fullana, A. Ribeiro, Constrained reinforcement learning has zero duality gap, in: *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- [31] J. Achiam, D. Held, A. Tamar, P. Abbeel, Constrained policy optimization, in: *Proc. International Conf. Machine Learning (ICML)*, 2017, pp. 22–31.
- [32] Y. Chow, O. Nachum, E. Duenez-Guzman, M. Ghavamzadeh, A Lyapunov-based approach to safe reinforcement learning, in: *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.
- [33] B. Könighofer, R. Bloem, N. Jansen, S. Junges, S. Pranger, Shields for safe reinforcement learning, *Communications of the ACM* 68 (11) (2025) 80–90. doi:10.1145/3715958.
- [34] V. V. Menon, H. Amirpour, M. Ghanbari, C. Timmerer, Perceptually-aware per-title encoding for adaptive video streaming, in: *Proc. IEEE International Conference on Multimedia and Expo (ICME)*, 2022, pp. 1–6. doi:10.1109/ICME52920.2022.9859744.
- [35] Federal Communications Commission, *Measuring broadband america - fixed broadband reports*, <https://www.fcc.gov/reports-research/reports/measuring-broadband-america>, ongoing program; data used for training and evaluation (2015).
- [36] A. Kvalbein, D. Baltrūnas, K. Evensen, J. Xiang, A. Elmokashfi, S. Ferlin, The nornet edge platform for mobile broadband measurements, *Computer Networks* 61 (2014) 88–101.
- [37] M. J. Neely, *Stochastic Network Optimization with Application to Communication and Queueing Systems*, Morgan & Claypool, 2010.
- [38] R. F. Barber, E. J. Candès, A. Ramdas, R. J. Tibshirani, Conformal prediction beyond exchangeability, *The Annals of Statistics* 51 (2) (2023) 816–845. doi:10.1214/23-AOS2276.
- [39] A. Raffin, A. Hill, M. Ernestus, A. Gleave, A. Kanervisto, N. Dormann, Stable-Baselines3: Reliable reinforcement learning implementations, *Journal of Machine Learning Research* 22 (268) (2021) 1–8.

- [40] K. Spiteri, R. K. Sitaraman, D. Sparacio, From theory to practice: Improving bitrate adaptation in the DASH reference player, in: Proc. ACM Multimedia Systems Conference (MMSys), 2018, pp. 123–137. doi : 10 . 1145/3204949 . 3204953.
- [41] D. Lakens, Equivalence tests: A practical primer for t tests, correlations, and meta-analyses, *Social Psychological and Personality Science* 8 (4) (2017) 355–362. doi : 10 . 1177/1948550617697177.
- [42] R. T. Rockafellar, S. Uryasev, Optimization of conditional value-at-risk, *Journal of Risk* 2 (2000) 21–42.